

Spiral Chart

The ***Square of Nine*** (originally called the ***Spiral Chart***) is a tool of [technical analysis](#) which was first popularized by famous market forecaster [W. D. Gann](#).^{[1][2][3]} The chart first appeared in his correspondence course *Master Charts*^[4], which is now a part of the *W. D. Gann Master Commodities Course*^[5]. It is very similar to the Hexagon Chart^[3], another chart described in the *Master Charts* course.

It is worth noting that Gann's original use of the term "Square of Nine" refers to a completely different chart, and he referred to what is now known as the Square of Nine today as the "Spiral Chart"^[4]. However, for some reason, later authors mixed up the terms and the name "Spiral Chart" has become much less popular today^[2].

Construction

The Square of Nine consists of the integers arranged in the form of a square spiral.^[6] One can construct it using a spreadsheet or graph paper. Place the number 1 in a central cell; to its left place the number 2; below that is 3, to the right of 3 is 4 and 5, then above 5 is 6 and then 7. To the left of 7 is 8 and then 9, which completes the first square. To the left of 9 place 10 to start a new square out, and below 10 place 11, and so on. Below is a simple example going up to 81:

81	80	79	78	77	76	75	74	73
50	49	48	47	46	45	44	43	72
51	26	25	24	23	22	21	42	71
52	27	10	9	8	7	20	41	70
53	28	11	2	1	6	19	40	69
54	29	12	3	4	5	18	39	68
55	30	13	14	15	16	17	38	67
56	31	32	33	34	35	36	37	66
57	58	59	60	61	62	63	64	65

Spiral direction

Regarding the direction of the spiral, it appears clockwise in some books^[3] and counterclockwise in others^[7]. In the original chart used by Gann, the direction was counterclockwise^[1]. Patrick Mikula explains that it is best to spin the numbers counterclockwise, because, onto the Square of Nine, Gann liked to overlay astrological charts, most of which spin counterclockwise^[7].

Angular value of each number

Zodiacal relationship in the Square of Nine

In the first volume of *Gann Scientific Methods Unveiled*^[7], Patrick Mikula assigns the following method to overly the planetary positions on the chart:

- The axis going from 1 to 2,11, 28, and so on are 0° or 0° [Aries](#), the [spring equinox](#) (in the [Northern Hemisphere](#), same for below).
- The axis going from 1 to 4, 15, 34 and so on are 90° or 0° [Cancer](#), the [summer solstice](#).
- The axis going from 1 to 6, 19, 40, 69 and so on are 180° or 0° [Libra](#), the [autumn equinox](#).
- The axis going from 1 to 8, 23, 46, 77, etc. is 270° or 0° [Capricorn](#), the [winter solstice](#).

Simplified form and universal fomula

In *Predicting Market Trends with Periodic Number Cycles*^[8], Carl A. Futia simplified the Square of Nine into a form which he calls the Octagon Chart. Basically, the Octagon Chart only records the numbers on the Square of Nine lying on the angles of the 8th harmonic. Here is an example:

	I	II	III	IV	V	VI	VII	VIII	IX
0°	2	11	28	53	86	127	176	233	298
45°	3	13	31	57	91	133	183	241	307
90°	4	15	34	61	96	139	190	249	316
135°	5	17	37	65	101	145	197	257	325
180°	6	19	40	69	106	151	204	265	334
225°	7	21	43	73	111	157	211	273	343
270°	8	23	46	77	116	163	218	281	352
315°	9	25	49	81	121	169	225	289	361

To calculate any number in between these angles, Futia^[8] provided a simple formula, which is described in the following table:

Futia's Square of Nine Formula, Explained Step by Step		
Step	Description	Operation
1	Let P be the number of which the angle you are trying to find.	P
2	Minus 1 from P	P−1
3	Take the square root of (P−1).	(P−1) ^½
4	Multiply (P−1) ^½ by 180, and then subtract 225 from it.	180×(P−1) ^½ −225
5	Divide the answer is Step 4 by 360 and find the remainder.	MOD[180×(P−1) ^½ −225,360]

In all: for any natural number P>1, its approximate angular value A (in degrees) is given by:

A = MOD[180×(P−1)^½−225,360]

For example, from the Octagon Chart, it is known that the angular value of 73 is 225°. Substitute P with 73 into the above formula:

A₇₃ = MOD[180×(73−1)^½−225,360] = MOD[180×8.4853−225,360] = 222.35, which is close to 225.

In [Microsoft Excel](#), if the value of P is input in the cell A1, then the formula to calculate the angle is MOD(SQRT(A1)*180-225,360). In [OpenOffice Calc](#), the same formula could be used except that the comma has to be replaced by a colon. Below are some examples using the formula in Excel:

	A	B	(The formula in B)
1	10	344.2099788	<-- MOD(SQRT(A1)*180-225,360)
2	20	219.9844719	<-- MOD(SQRT(A2)*180-225,360)
3	30	40.90060351	<-- MOD(SQRT(A3)*180-225,360)

Angular locations of perfect squares

As easily seen from the above numbers, the odd squares (9, 25, 49, etc.) progress along the 315° line on the chart. Conversely, the even squares move closely along the 45° line, which is in the exact opposite direction. This is different from the [Hexagon Chart](#), when the perfect squares are always 210° away from each other.

References

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